



Karnataka State Police
Government of Karnataka

Powered by **H2S**

0atathon 2026

Technology partner



Team Details

- a. Team name: ARP Tech
- b. Team leader name: RaviTeja M
- c. Team size: 4
- d. Problem Statement: Intelligent Conversational AI for KSP Crime Database



Brief about the solution :

Intelligent Conversational AI for Crime Database Analytics lets officers, analysts, and administrators query crime databases using plain language instead of SQL.

How it works: User asks a question → AI understands intent → generates a safe, read-only SQL query → runs it against the database → returns a natural-language summary + chart/table/map, with the query shown for transparency.

Example: *"Show theft cases in Sector 4, last 3 months, sorted by severity"* → 47 cases found, auto-generated chart shows a 32% spike.

Key strengths:

- No SQL knowledge needed
- Supports follow-up questions (conversational memory)
- Auto-visualizes results (charts, maps, hotspots)
- Secure by design — read-only, role-based access, every query auditable



Opportunities :

a. How different is it from any of the other existing ideas?

Most existing crime-data tools fall into two categories: **rigid dashboards** (pre-built reports/filters that can't answer anything outside their fixed templates) or **generic BI tools** (Power BI/Tableau — powerful, but still need someone technical to build the queries and visuals first).

a. How will it be able to solve the problem?

- Removes the technical bottleneck
- Speeds up pattern
- Enables real-time, in-field
- Reduces human error
- Scales access without scaling

a. USP of the proposed solution

- Conversational simplicity.
- Transparent by default.
- Security-native design
- Action-oriented, not just informative



List of features offered by the solution:

- Natural Language to SQL (Text-to-SQL)
- Conversational Memory
- Automatic Visualization
- Geospatial Crime Mapping
- Trend & Anomaly Detection
- Role-Based Access Control (RBAC)
- PII Masking
- Explainability / Query
- Read-Only Query Guardrails
- Multilingual & Voice Input
- Multi-Modal Output
- Audit Logging



Process flow diagram or Use-case diagram :



Officer



Analyst

Crime analytics AI system

Ask natural language query

View charts, tables & maps

Get trend & anomaly alerts

Refine with follow-ups

Manage roles & access

Audit generated queries



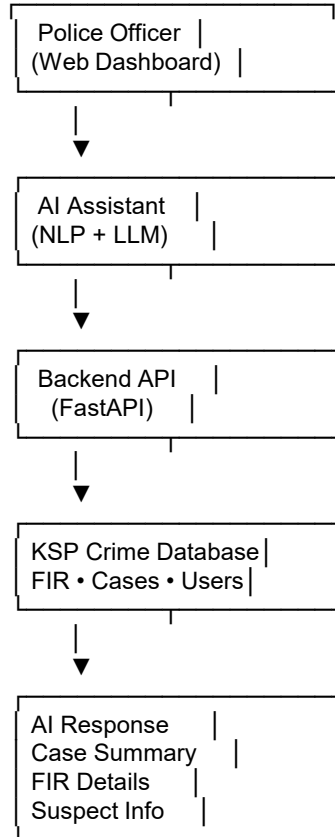
Administrator



Public user



Architecture diagram of the proposed solution





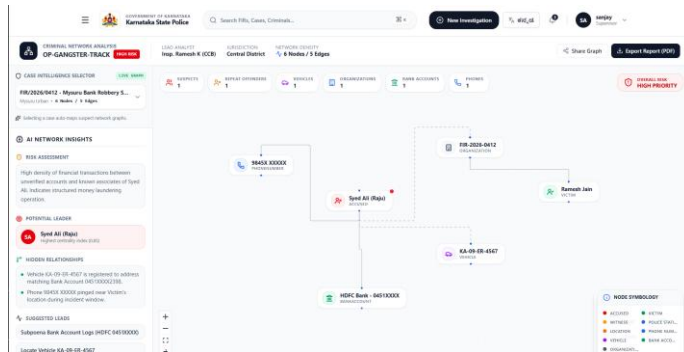
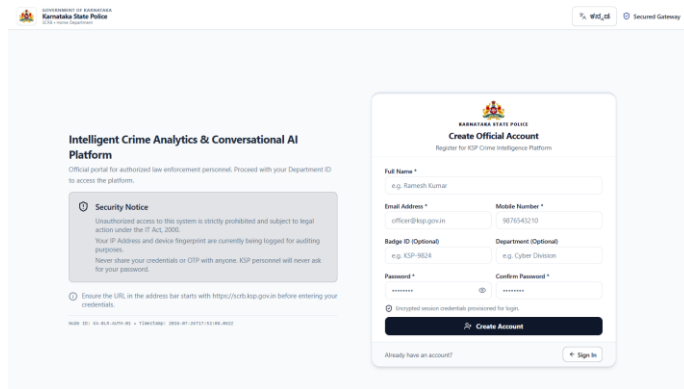
Technologies to be used in the solution

- ❖ **Frontend:** Next.js (React), TailwindCSS
- ❖ **Backend:** FastAPI (Python)
- ❖ **Database:** Zoho Catalyst Data Store
- ❖ **Authentication:** Zoho CatalystAuth(JWT)
- ❖ **AI:** Zoho Catalyst AI (Zia) / LLM
- ❖ **Testing:** Vitest
- ❖ **Deployment:** Zoho Catalyst / Vercel
- ❖ **Version Control:** Git & GitHub



List down Catalyst Services being used in the solution

- Catalyst Authentication
- Catalyst Data Store
- Catalyst Functions
- Catalyst AI (Zia)
- Catalyst API Gateway
- Catalyst Hosting
- Catalyst Logs





Prototype Performance report/Benchmarking

- Fast and secure authentication for police officers.
- AI processes natural language queries in a few seconds.
- Quick retrieval of FIRs, case records, and suspect information.
- High accuracy in fetching relevant case data.
- Scalable architecture for future expansion across police departments.



Provide links to your:

1. **GitHub Public Repository** : <https://github.com/ravitejam2007-code/karnataka-state-police.git>
2. **Deployed Link** : <https://ksp-arptech.onslate.in>



Karnataka State Police
Government of Karnataka

Technology partner



Powered by

I12S
BACKSTILL



GOVERNMENT OF KARNATAKA
Karnataka State Police
State Crime Records Bureau

[Home](#)

[Citizen Services](#)

[News & Updates](#)

[Contact](#)

ಕರ್ನಾಟಕ ಪೊಲೀಸ್

[Citizen Login](#)

Supporting Investigation • Crime Analytics • Predictive Intelligence • Secure Operations

 CITIZEN SERVICES

Online Citizen Services

Access a wide range of police services online from the comfort of your home. No need to visit the police station for these services.



E-FIR Registration

Register a complaint online and track FIR status in real-time



Missing Person Search

Report or search for missing persons across Karnataka



Vehicle Theft Report

Register stolen vehicle complaints and check recovery status



Character Verification

Apply for police clearance certificate and background verification



Karnataka State Police
Government of Karnataka

Powered by **1125**

Datathon

2026

Technology partner



THANK YOU